

Baisani Narasimha Sai Manas

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SKILLS

- **Languages:** C, C++, Java, Embedded C, Verilog HDL
- **Microcontrollers:** ESP32, Arduino, 8051, PIC P1F458, STM 32
- **Tools/Platforms:** Xilinx Vivado, Cadence NC-Sim, Proteus, Cisco Packet Tracer
- **Soft Skills:** Adaptability, Collaborative, Problem Solving

PROJECTS

- **Secure Wireless Encrypted Data Logging System** Apr' 26
 - Architected a dual-node embedded framework utilizing ESP32 microcontrollers and NRF24L01 transceivers for secure wireless acquisition and communication of sensor data.
 - Integrated lightweight cryptographic processing, DS3231 RTC-based timestamping, LCD visualization, and microSD storage to establish a secure end-to-end embedded data logging pipeline.
 - Leveraged SPI, I2C, GPIO, and RF communication interfaces to enable reliable inter-node data transfer and modular storage for scalable IoT and industrial monitoring applications.**Tech: Embedded C, ESP32, NRF24L01, SPI, I2C, GPIO, DS3231 RTC, microSD, LCD, DHT11**
- **IoT-Based Smart Door Access Control System** Dec' 25
 - Established a time-based smart door access framework using ESP32, RTC, and servo actuators to enable secure scheduled entry with real-time operation.
 - Integrated a mobile application interface for remote configuration of access schedules while supporting push-button authentication and servo-driven locking mechanisms.
 - Extended the system architecture to support multi-door access management and centralized monitoring, improving system scalability and operational flexibility.**Tech: Embedded C, MicroPython, ESP32, DS3231 RTC, Servo Motors, MIT App Inventor**
- **8-Bit Single-Cycle RISC Processor Design and RTL Verification** Nov' 25
 - Architected a custom 8-bit RISC processor from scratch using Verilog HDL, implementing a single-cycle Harvard architecture with a 16-bit instruction set and an 8-register file.
 - Achieved a high-performance ALU and a centralized Control Unit capable of decoding and executing R-type, I-type, and Branch instructions while managing status flags for conditional logic.
 - Validated full-system functionality through RTL simulation in Vivado, developing comprehensive testbenches and assembly programs to verify data memory operations, branch accuracy, and register file integrity.**Tech: Verilog HDL, Digital Electronics, RTL Design, Vivado**

TRAINING

- **Digital System Design | Trainee | [Certificate](#)** Jul' 25
 - Gained strong foundation in combinational and sequential circuits.
 - Implemented different types of counters, shift registers on Proteus Software.
 - Explored the architecture of different types of Static-RAMS and Dynamic-RAMS.

CERTIFICATES

- Mastering Microcontroller and Embedded Driver Development | Udemy | [Certificate](#) Mar' 26
- Microcontroller Embedded C Programming: Absolute Beginners | Udemy | [Certificate](#) Feb' 25
- Hardware Modeling using Verilog | NPTEL | [Certificate](#) Apr' 25
- Microprocessors and Interfacing | NPTEL | [Certificate](#) Dec' 24

ACHIEVEMENTS

- N5 certified by Centre for Foreign Language LPU Jan' 25
- First runner up in 'PROJECT-O-MANIA'. Jan' 24

EDUCATION

Lovely Professional University	Punjab, India
Bachelor of Technology - Electronics and Communication Engineering: CGPA- 8.23	Since Aug' 23
Narayana Junior College	Vijayawada, Andhra Pradesh
Intermediate; Percentage: 95.3	Jun' 21 - Mar' 23
Viswabharati High School	Gudivada, Andhra Pradesh
Matriculation; Percentage: 100	Jul' 20 - Apr' 21